Results Analysis

```
import pandas as pd
import os
import matplotlib.pyplot as plt
import seaborn as sns
from scipy.stats import shapiro
import scipy.stats as stats
```

Obtaining datasets

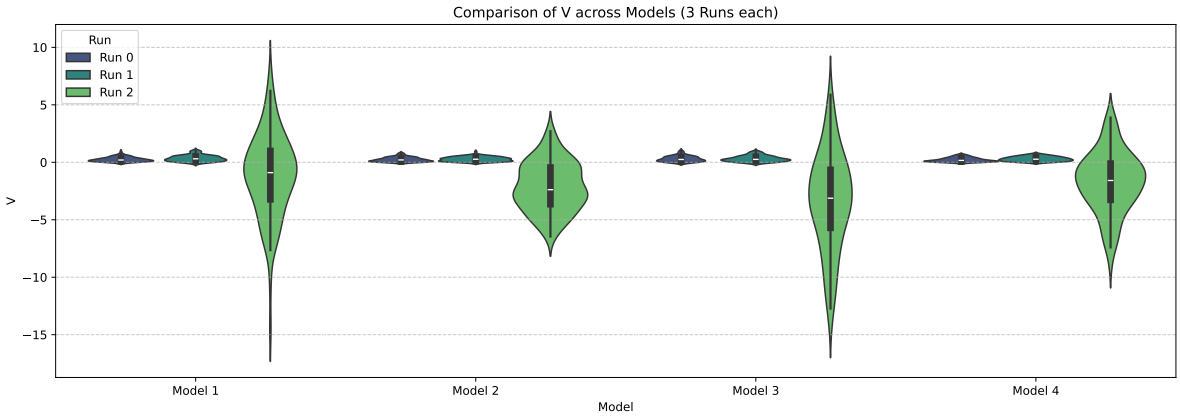
```
lvdsl_output=[
  ## Primer modelo
  "./lvdsl_outputs/VacuaFound_07052026_174039/GA/GA-0000.csv.csv",
  "./lvdsl_outputs/VacuaFound_07052026_174039/GA/GA-0001.csv.csv",
  "./lvdsl_outputs/VacuaFound_07052026_174039/GA/GA-0002.csv.csv",
  ## Segundo modelo
  "./lvdsl_outputs/VacuaFound_07052026_174049/GA/GA-0000.csv.csv",
  "./lvdsl_outputs/VacuaFound_07052026_174049/GA/GA-0001.csv.csv",
  "./lvdsl_outputs/VacuaFound_07052026_174049/GA/GA-0002.csv.csv",
  ## Tercer modelo
  "./lvdsl_outputs/VacuaFound_07052026_174804/GA/GA-0000.csv.csv",
  "./lvdsl_outputs/VacuaFound_07052026_174804/GA/GA-0001.csv.csv",
  "./lvdsl_outputs/VacuaFound_07052026_174804/GA/GA-0002.csv.csv",
  ## Cuarto modelo
  "./lvdsl_outputs/VacuaFound_07052026_175255/GA/GA-0000.csv.csv",
  "./lvdsl_outputs/VacuaFound_07052026_175255/GA/GA-0001.csv.csv",
  "./lvdsl_outputs/VacuaFound_07052026_175255/GA/GA-0002.csv.csv",
]

data=[]

for dataset in lvdsl_output:
  data.append(pd.read_csv(dataset))
```

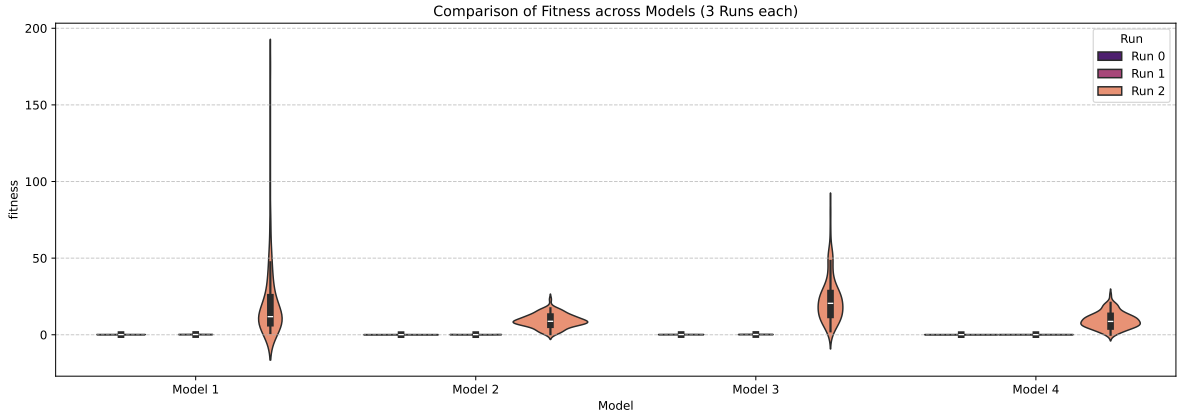
by Model grouping

```
all_frames = []

for i, path in enumerate(lvdsl_output):
    df = pd.read_csv(path)
    # Assign model group (0-2 is Model 1, 3-5 is Model 2, etc.)
    df['Model'] = f'Model {i // 3 + 1}'
    # Assign Run ID (0, 1, or 2 within that model)
    df['Run'] = f'Run {i % 3}'
    all_frames.append(df[['V', 'fitness', 'Model', 'Run']])
```

Fitness and Potential comparison

```
# Combine into one master dataframe
combined_df = pd.concat(all_frames)

# 2. Create the Comparison Plot
fig, axs = plt.subplots(2, 1, figsize=(14, 10))

# Plot for V
sns.violinplot(data=combined_df, x='Model', y='V', hue='Run', ax=axs[0], palette='viridis')
axs[0].set_title('Comparison of V across Models (3 Runs each)')
axs[0].grid(axis='y', linestyle='--', alpha=0.7)

# Plot for fitness
sns.violinplot(data=combined_df, x='Model', y='fitness', hue='Run', ax=axs[1], palette='magma
axs[1].set_title('Comparison of Fitness across Models (3 Runs each)')
axs[1].grid(axis='y', linestyle='--', alpha=0.7)

plt.tight_layout()
plt.show()
```





Normality test

```
normality_results = []

for i, df in enumerate(data):
    # Test for 'V'
    stat_v, p_v = shapiro(df['V'])
    # Test for 'fitness'
    stat_f, p_f = shapiro(df['fitness'])

    normality_results.append({
        'Dataset': f'DF {i}',
        'V_p-value': round(p_v, 4),
        'V_is_Normal': p_v > 0.05,
        'Fit_p-value': round(p_f, 4),
        'Fit_is_Normal': p_f > 0.05
    })
```

```
# Display as a Table
results_df = pd.DataFrame(normality_results)

#plt.figure(figsize=(6, 6))
#stats.probplot(data[0]['fitness'], dist="norm", plot=plt)
#plt.title("Q-Q Plot for Fitness (Dataset 0)")
#plt.show()

print(results_df)
```

```
   Dataset  V_p-value  V_is_Normal  Fit_p-value  Fit_is_Normal
0     DF 0     0.0001        False       0.1046           True
1     DF 1     0.0007        False       0.0000          False
2     DF 2     0.1623         True       0.0000          False
3     DF 3     0.0000        False       0.3620           True
4     DF 4     0.0036        False       0.0719           True
5     DF 5     0.3750         True       0.3386           True
6     DF 6     0.0001        False       0.5932           True
7     DF 7     0.0003        False       0.0248          False
8     DF 8     0.3651         True       0.0000          False
9     DF 9     0.0000        False       0.6264           True
10   DF 10     0.0042        False       0.2522           True
11   DF 11     0.3894         True       0.0284          False
```

Tests

```
from scipy.stats import shapiro, f_oneway, kruskal

# Defining your matrix groups
groups = [
    [0, 3, 6, 9],
    [1, 4, 7, 10],
    [2, 5, 8, 11]
]

target_metrics = ['V', 'fitness']

for row_idx, group in enumerate(groups):
    # Extract s and tau from the first dataframe in the group
    s_val = data[group[0]]['s'].iloc[0]
    tau_val = data[group[0]]['tau'].iloc[0]
```

```
    print(f"--- Testing Moduli Group {row_idx}: s={s_val}, tau={tau_val} ---")

    for metric in target_metrics:
        # Collect the metric arrays for the 4 dataframes in this group
        samples = [data[i][metric] for i in group]

        # 1. Normality Check (All samples in group must be normal for ANOVA)
        all_normal = all(shapiro(s)[1] > 0.05 for s in samples)

        # 2. Perform Test
        if all_normal:
            stat, p_value = f_oneway(*samples)
            test_name = "ANOVA"
        else:
            stat, p_value = kruskal(*samples)
            test_name = "Kruskal-Wallis"

        result = "Significant Difference" if p_value < 0.05 else "No Significant Difference"

        print(f"  [{metric}] {test_name}: p={p_value:.4f} -> {result}")
    print("\n")
```

```
--- Testing Moduli Group 0: s=1.600149459561681, tau=3.485910062937601 ---
  [V] Kruskal-Wallis: p=0.1795 -> No Significant Difference
  [fitness] ANOVA: p=0.0000 -> Significant Difference


--- Testing Moduli Group 1: s=2.459242111292193, tau=3.0947272010640288 ---
  [V] Kruskal-Wallis: p=0.5771 -> No Significant Difference
  [fitness] Kruskal-Wallis: p=0.0000 -> Significant Difference


--- Testing Moduli Group 2: s=1.053781742958659, tau=1.4533862399266009 ---
  [V] ANOVA: p=0.0000 -> Significant Difference
  [fitness] Kruskal-Wallis: p=0.0000 -> Significant Difference
```

Posthoc

```python
from scipy.stats import mannwhitneyu

# Your matrix again
groups = [[0, 3, 6, 9], [1, 4, 7, 10], [2, 5, 8, 11]]
model_names = ["Model 1", "Model 2", "Model 3", "Model 4"]

for row_idx, group_indices in enumerate(groups):
    print(f"--- ANALYZING MODULI GROUP {row_idx} ---")

    # 1. Calculate Medians to find the 'Candidate' for best
    medians = [data[i]['fitness'].median() for i in group_indices]
    best_idx_in_group = medians.index(max(medians))
    best_model_name = model_names[best_idx_in_group]
    best_data = data[group_indices[best_idx_in_group]]['fitness']

    print(f"Candidate for Best: {best_model_name} (Median Fitness: {max(medians):.4f})")

    # 2. Pairwise Comparison (Best vs Everyone Else)
    # Since we do 3 comparisons per group, alpha becomes 0.05 / 3 = 0.0166 (Bonferroni)
    alpha_corrected = 0.05 / 3
    is_truly_superior = True

    for i, model_idx in enumerate(group_indices):
        if i == best_idx_in_group:
            continue

        compare_data = data[model_idx]['fitness']
        stat, p = mannwhitneyu(best_data, compare_data, alternative='greater')

        if p > alpha_corrected:
            print(f"  - No statistical lead over {model_names[i]} (p={p:.4f})")
            is_truly_superior = False
        else:
            print(f"  - Statistically better than {model_names[i]} (p={p:.4f})")

    if is_truly_superior:
        print(f"RESULT: {best_model_name} is the CLEAR WINNER for this group.\n")
    else:
        print(f"RESULT: Tie or negligible difference between top models.\n")
```

```
--- ANALYZING MODULI GROUP 0 ---
Candidate for Best: Model 3 (Median Fitness: 0.1215)
```

```
  - Statistically better than Model 1 (p=0.0044)
  - Statistically better than Model 2 (p=0.0000)
  - Statistically better than Model 4 (p=0.0000)
RESULT: Model 3 is the CLEAR WINNER for this group.

--- ANALYZING MODULI GROUP 1 ---
Candidate for Best: Model 3 (Median Fitness: 0.1560)
  - No statistical lead over Model 1 (p=0.0552)
  - Statistically better than Model 2 (p=0.0000)
  - Statistically better than Model 4 (p=0.0000)
RESULT: Tie or negligible difference between top models.

--- ANALYZING MODULI GROUP 2 ---
Candidate for Best: Model 3 (Median Fitness: 20.4978)
  - Statistically better than Model 1 (p=0.0002)
  - Statistically better than Model 2 (p=0.0000)
  - Statistically better than Model 4 (p=0.0000)
RESULT: Model 3 is the CLEAR WINNER for this group.
```